

Indian Institute of Technology Madras

DA5400W - Foundations of machine learning

Assignment II

Due date: February 15, 2026, 12 pm

Instruction

1. Assignment shall be submitted on the due date. Late submissions will not be entertained. Any submission through an email will have a -20 mark penalty.
2. All the assignments must be the student's own work.

Question 1:

Consider a dataset with two numerical features:

- x_1 measured in kilometers, with values approximately ranging from 0 to 10.
- x_2 measured in meters, with values approximately ranging from 0 to 500.

No preprocessing or scaling is applied to the data.

- (a) Explain how the difference in feature scales affects clustering when Euclidean distance is used.
- (b) If K-means clustering is applied directly to this dataset without preprocessing, which feature is likely to dominate the clustering result? Justify your answer.
- (c) Describe one preprocessing technique to address this issue and explain how it changes the geometry of the data.

Question 2:

- (a) Explain the role of distance (or similarity) measures in clustering algorithms.
- (b) Compare **Euclidean distance** and **Mahalanobis distance**. Discuss how the choice of distance measure affects:
 - The shape of clusters
 - Sensitivity to feature scaling and correlation
- (c) Give one practical scenario where Euclidean distance may lead to poor clustering results and justify your answer.

Question 3:

- (a) Explain the intuition behind the Knee (Elbow) Method for selecting the number of clusters in partitional clustering algorithms.
- (b) Discuss two limitations of the Knee Method when applied to real-world datasets.
- (c) Describe one alternative approach for estimating the number of clusters and explain how it differs conceptually from the Knee Method.

Question 4:

- (a) Explain the working principle of hierarchical agglomerative clustering. What is the role of a dendrogram?
- (b) Compare the following linkage criteria:
 - Single-link
 - Complete-link
 - Average-link
 in terms of cluster shape, sensitivity to noise, and tendency to form compact clusters.
- (c) Explain how the choice of linkage affects the structure of the resulting dendrogram.

Question 5:

Hierarchical agglomerative clustering is applied to a dataset using single-linkage clustering.

- (a) Explain how the presence of a small number of noisy or outlier points can significantly alter the resulting dendrogram.
- (b) Compare the sensitivity of single-linkage and complete-linkage clustering to such noise.
- (c) Discuss how this sensitivity impacts the interpretability of clusters obtained by cutting the dendrogram at a fixed height.

Question 6:

- (a) Explain how a dataset can be represented as a weighted undirected graph for clustering purposes.
- (b) Describe how single-link and complete-link hierarchical clustering can be interpreted using thresholded graphs.
- (c) Using this graph-theoretic view, discuss one strength and one weakness of each method.

Question 7:

The k -means algorithm (Lloyd's algorithm) is widely used for partitional clustering.

- (a) Can the algorithm result in fewer than k clusters at any iteration, even if k was initially specified? Justify your answer.
- (b) Can the algorithm ever return to the same clustering arrangement that it had in any of the previous iterations? Justify your answer.

Question 8:

Consider the following points:

$$x_1 = [0, -1], \quad x_2 = [0, 1], \quad x_3 = [1, 0], \quad x_4 = [1, 0], \quad x_5 = [-1, 0], \quad x_6 = [0, 0].$$

For $k = 3$, is $\{x_3, x_6, x_4\}$ a valid order of means selected during k -means++ initialization? If not, suggest one valid possible order and justify briefly.

Question 9:

Consider the following 2D data points:

$$\begin{aligned} P_1 &: (7.0, 6.5), & P_2 &: (5.0, 10.0), \\ P_3 &: (6.2, 7.1), & P_4 &: (2.0, 3.1), \\ P_5 &: (9.3, 2.4), & P_6 &: (8.5, 1.9), \\ P_7 &: (2.8, 3.6). \end{aligned}$$

Apply the DBSCAN algorithm with parameters $\text{minPts} = 2$ and $\varepsilon = 1.15$ using Euclidean distance.

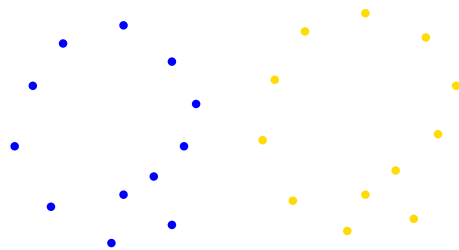
After running DBSCAN with the given parameters, determine the number of clusters formed.

Question 10:

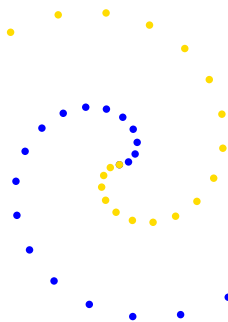
For each of the following datasets (a)–(d), assume the number of clusters is $k = 2$. Which clustering method among the following would work best? Justify briefly.

- Hierarchical clustering with single-link
- Hierarchical clustering with complete-link
- Hierarchical clustering with average-link
- k -means
- Gaussian Mixture Model (with no restriction on covariance matrices)

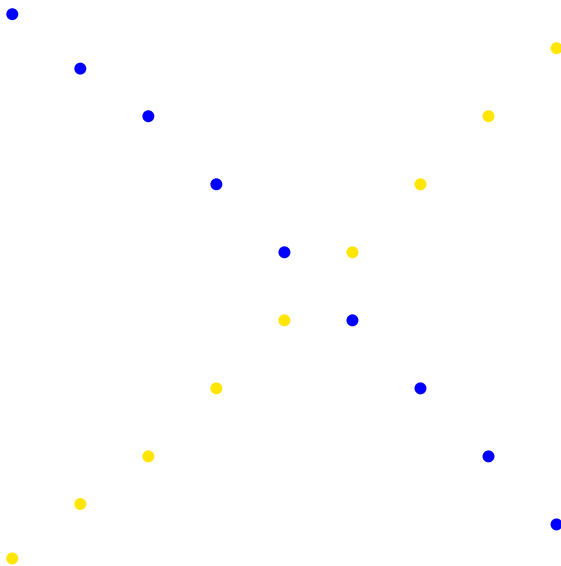
(a)



(b)



(c)



(d)

